MarxKapital: A Computational Framework for Marxian Value Theory with Real Empirical Calibration

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July 2026

Abstract

We present MarxKapital, an open-source Julia package for the mathematical modeling of Marxian value theory: capital accumulation, the tendency of the rate of profit, the transformation of labor values into prices of production, and the sectoral dynamics of expanded reproduction. Beyond the classical two-department and generalized N-department reproduction schemes, the package implements three complementary mechanisms for coupling sectoral accumulation through the technical matrix A — capital mobility, an aggregate realization-pressure heuristic, and an exact per-sector quantity-rationing scheme solved by fixed-point iteration — without recourse to differential-algebraic equation (DAE) solvers. We further present a live, reproducible empirical pipeline that calibrates the model directly from official European statistical sources (Eurostat and AMECO, via their SDMX REST APIs), replacing the synthetic or ad hoc calibration common in the computational Marxian literature. Applying this pipeline to Portugal, we document a sectoral profile for 2022 and an aggregate profit-rate series for 1996–2022 in which the rate of exploitation rises markedly while the organic composition of capital falls, yielding a rising profit rate over the period — a finding we discuss critically rather than as a settled conclusion, given the measurement conventions and simplifications involved.

1 Introduction

Empirical and computational work in the Marxian tradition faces a recurring reproducibility problem: profit-rate and organic-composition estimates in the literature are frequently produced by bespoke, non-public spreadsheets or scripts, calibrated to data sources whose vintage and exact construction are difficult to reproduce [Shaikh, 2016, Duménil and Lévy, 1993]. At the same time, existing open computational implementations of Marxian reproduction schemes tend to rely on synthetic or randomly generated sectoral data for demonstration purposes, since assembling a real technical (input-output) matrix, real sectoral value-added and employment series, and real capital stock data from official sources is itself a substantial undertaking.

MarxKapital is an attempt to close both gaps at once. It is, first, a methodological contribution: a Julia implementation of the classical two-department reproduction scheme, its generalization to N departments, the labor-value and Sraffian production-price apparatus for the transformation problem, and — the paper's central technical contribution — three distinct mechanisms for coupling N-department accumulation dynamics through the technical matrix, culminating in an exact quantity-rationing scheme in the spirit of the disequilibrium-macro literature [Barro and Grossman, 1971, Bénassy, 1975], implemented without the numerical machinery a literal differential-algebraic formulation would require. It is, second, an empirical contribution: a live data pipeline that fetches real sectoral national accounts, input-output tables, and macroeconomic series from Eurostat and the European Commission's AMECO database via their official SDMX REST APIs, and uses them to calibrate every quantity the model needs — constant and

variable capital, the rate of exploitation, the organic composition, the technical matrix itself, and (optionally) a perpetual-inventory capital-stock series — with no synthetic fallback unless explicitly requested.

We use this pipeline to calibrate the model for Portugal and report two sets of results: a detailed 19-sector profile for 2022, and an aggregate time series of the rate of exploitation, organic composition, and profit rate for 1996–2022. The latter shows a profit rate that rises over the period, driven by a rate of exploitation that rises faster than the organic composition falls. We present this as a genuine empirical finding from real, unmodified official data, but discuss it critically: it runs against the classical tendential-fall expectation, and we consider several candidate explanations without asserting any of them as established.

The remainder of the paper is organized as follows. Section 2 situates the model in the relevant theoretical literature. Section 3 presents the model itself, including the coupled N-department dynamics and the quantity-rationing mechanism. Section 4 describes the empirical calibration pipeline and the Portuguese data. Section 5 reports results. Section 6 discusses the findings and the model's limitations. Section 7 concludes.

2 Related work

The theoretical foundations implemented here are classical: the reproduction schemes of Marx [1976], the tendency of the rate of profit discussed in Marx [1981], and the transformation of values into prices of production, for which we adopt the Sraffian linear-production-price formulation of Sraffa [1960]. Okishio [1961] established the conditions under which technical change and the choice of technique affect the profit rate, a result the package's technical-matrix machinery is built to be consistent with, though we do not implement an explicit choice-of-technique search. On the empirical side, Shaikh [2016] and Duménil and Lévy [1993] are the standard references for long-run profit-rate estimation from national accounts; our approach differs mainly in emphasizing a fully automated, re-runnable data pipeline over manual series construction. The exact quantity-rationing mechanism introduced in Section 3.3 draws on the disequilibrium-macroeconomics literature on short-side rationing rules, in particular Barro and Grossman [1971] and Bénassy [1975], applied here to a Leontief-technology setting rather than the labor/goods markets those papers originally analyze.

3 The model

3.1 The classical two-department scheme

The package's starting point is Marx's classical two-department scheme: Department I produces means of production, Department II produces consumption goods. Each department $d \in \{1, 2\}$ is characterized by constant capital C_d, variable capital V_d, and a rate of exploitation e_d, giving surplus value $S_d = e_d V_d$, organic composition $k_d = C_d/V_d$, and profit rate $r_d = e_d/(k_d + 1)$. Accumulation follows

$$\dot{C}_d = \sigma_{c,d} S_d - \delta_d C_d, \qquad \dot{V}_d = \sigma_{v,d} S_d, \tag{1}$$

where $\sigma_{c,d} + \sigma_{v,d} + \gamma_d = 1$ splits surplus value between reinvestment in constant capital, reinvestment in variable capital, and capitalist consumption, and δ_d is a depreciation rate. Intersectoral consistency (does Department I's wage and unaccumulated-surplus income suffice to buy Department II's output, and vice versa) is checked *a posteriori* by a dedicated consistency function, not imposed inside the differential system — a design choice the N-department generalization inherits and, in one respect, improves upon (Section 3.3).

3.2 Generalization to N departments and the technical matrix

For N sectors, the model introduces a technical matrix $A \in \mathbb{R}_{\geq 0}^{N \times N}$, with A_{ij} the value of good i required as input per unit of output of sector j, constructed either synthetically (with structural options for random, triangular, or block sparsity patterns, always rescaled to spectral radius below unity) or — our empirical contribution — from a real input-output table (Section 4). Given A and a labor-coefficient vector l, labor values follow the Leontief inverse, $\lambda = (I - A)^{-1} l$, and Sraffian production prices follow

$$p = w(1 + \bar{r}) \left[I - (1 + \bar{r}) A^{\top} \right]^{-1} l, \tag{2}$$

valid for any uniform profit rate $\bar{r}$ below the maximum feasible rate $R_{\max} = 1/\lambda_{\max}(A^{\top}) - 1$, where $\lambda_{\max}$ denotes the dominant eigenvalue. The baseline N-department accumulation system treats sectors as independent, exactly mirroring the two-department equations sector by sector; A is available to the value/price apparatus above but does not, in this baseline system, feed back into the accumulation dynamics themselves.

3.3 Coupling sectors through A: three mechanisms

The baseline independence of sectors is a known limitation of purely value-accounting reproduction models: physically, one sector's constant capital is another sector's output, and the technical matrix should constrain how much investment a sector can realize, not merely how its value and price magnitudes are computed. MarxKapital implements three progressively more demanding mechanisms to address this, all opt-in and mutually composable except where noted, applied to a coupled system whose accumulation equation for constant capital becomes

$$\dot{C}_i = \kappa_i \, \sigma_{c,i} S_i - \delta_i C_i + \mu \, (r_i - \bar{r}) \, C_i, \tag{3}$$

where r_i is computed with constant capital valued at Sraffa production prices (using the current aggregate profit rate, clamped below $0.95\, R_{\max}$ to avoid the pricing singularity, with a labor-value fallback if the price system is ill-conditioned) rather than raw book value, so that A enters the dynamics through valuation as well as through the mechanisms below.

Capital mobility. The term $\mu(r_i - \bar{r})C_i$ reallocates investment toward above-average-profit sectors, a direct computational analogue of capital's tendency to flow toward higher returns.

Aggregate realization pressure. Setting $\kappa_i \equiv \min(\rho_{\text{real}}, 1)$ uniformly across sectors, with $\rho_{\text{real}} = \sum_i \min(O_i, (AO)_i) / \sum_i (AO)_i$ and gross output $O_i = C_i + V_i + S_i$, dampens planned reinvestment economy-wide whenever aggregate intermediate demand implied by A and current outputs exceeds what those same outputs can supply. This is a coarse, aggregate approximation — it does not distinguish which sectors are actually constrained — but requires no iterative solution.

Exact per-sector quantity rationing. The paper's central technical contribution replaces the aggregate heuristic with a genuine per-sector rationing scheme. Under Leontief technology, a sector's planned reinvestment cannot substitute a shortage of one input with more of another; realized investment is bounded by the scarcest required input, and a sector's own shortfall reduces its effective demand for *all* its inputs, which in turn affects the rationing faced by its suppliers. This makes the problem simultaneous. Writing $\rho_i \in (0, 1]$ for the rationing factor faced by *suppliers* of good i and κ_j for the realized fraction of planned reinvestment for *buyer* j, the fixed point is

$$\kappa_j = \min_{i:\, A_{ij} > 0} \rho_i, \qquad D_i = \sum_j A_{ij} \, \kappa_j \, \sigma_{c,j} S_j, \qquad \rho_i = \min\left(1, \frac{O_i}{D_i}\right). \tag{4}$$

We solve this by damped (under-relaxed) fixed-point iteration inside the right-hand side of the ODE at every solver evaluation, in exactly the same spirit as the Sraffa-price linear solve above, rather than exposing (ρ, κ) as differential-algebraic state to the solver. This choice is deliberate: a literal DAE formulation (mass-matrix or index reduction) or a general nonlinear-complementarity solver would add real numerical risk and a new package dependency for no benefit over the fixed point, which we verified converges reliably (a closed-form two-sector example with a known solution $\rho_1 = \sqrt{0.8}$, and no convergence failures across more than twenty parameter combinations at $N = 8$ and at $N = 19$). We emphasize that undamped iteration is *not* reliable here: we verified by hand that a small two-sector example oscillates indefinitely between two states without under-relaxation, a concrete instance of the well-known failure mode of naive Picard iteration on min-based maps.

None of the three mechanisms imposes exact market clearing at every instant for every sector as a hard constraint solved simultaneously with the differential state — that would require the DAE formulation we deliberately avoid. A dedicated consistency-check function generalizes the two-department *a posteriori* check to N sectors, reporting the per-sector gap $O_i - (AO)_i$ and the aggregate realization ratio at a given point in the trajectory, for diagnostic use outside the dynamics themselves.

4 Empirical calibration

4.1 Data sources

MarxKapital fetches four families of real sectoral and macroeconomic data [Eurostat, 2026b,a, European Commission, Directorate-General for Economic and Financial Affairs, 2026], each via an official SDMX REST API rather than a static file:

- **Sectoral value added, compensation, and consumption of fixed capital** from Eurostat's national accounts by industry (`NAMA_10_A64`), at the 19-section NACE Rev. 2 level.
- **Sectoral employment and gross fixed capital formation** from the companion Eurostat datasets `NAMA_10_A64_E` and `NAMA_10_A64_P5`.
- **A real technical matrix** from Eurostat's symmetric input-output table at basic prices (`NAIO_10_CP1620`), product by product, aggregated to Leontief technical coefficients weighted by sectoral value added.
- **Aggregate macroeconomic series** (GDP, employment, compensation, capital stock, and eighteen further variables) from the European Commission's AMECO database.

All four sources were, until recently, distributed as static bulk files with heuristic, undocumented formats; both institutions have since migrated to official SDMX REST APIs, which MarxKapital uses directly, with the legacy static-file parsers retained only as a manual fallback.

4.2 Constructing the technical matrix

The symmetric input-output table records trade and transport margins as part of national-accounting convention, which can produce negative entries in the trade and transport product rows when the raw flows are aggregated naively to a coarser sectoral grouping. We avoid this by building the technical matrix at the full 19-sector resolution first — where the margin artifacts are confined to individual cells and can be clipped to zero without discarding genuine flows from other sectors — and only then aggregating, when a coarser grouping is needed, using the standard Leontief aggregation formula (value-added-weighted averaging of already non-negative

coefficients) rather than re-aggregating raw flows. The resulting 19-sector matrix for Portugal is well behaved: non-negative throughout, with spectral radius $\rho(A) \approx 0.13$ (Figure 2), comfortably below unity.

4.3 Depreciation and capital stock

By default the model uses the net capital stock reported directly by Eurostat for the requested year. As an option, it instead constructs a perpetual-inventory series, $K_t = (1 - \delta)K_{t-1} + I_t$, seeded from the earliest year with both a reported stock and investment series, with δ estimated per sector from the ratio of consumption of fixed capital to stock in that base year rather than assumed. This uses a single depreciation rate per sector, not differentiated by asset type; we return to this simplification in Section 6.2.

5 Results: Portugal

5.1 Sectoral profile, 2022

Table 1 reports the calibrated rate of exploitation, organic composition, and profit rate for all 19 sections, ranked by profit rate. The value-added-weighted means are $e \approx 0.89$, $k \approx 24.8$, and $r \approx 10.0\%$; the mean organic composition is dominated by a single outlier, real estate activities ($k \approx 334$), which reflects the well-known effect of imputed owner-occupied rents inflating capital intensity in this sector's national-accounts treatment rather than a genuine production characteristic, and should be discounted accordingly when interpreting the sectoral mean. Excluding real estate, sectoral profit rates range from financial and insurance activities at the top ($r \approx 36\%$) to public administration and utilities near the bottom (r below 2%), a dispersion consistent with the classical expectation that market prices, not labor values, equalize profit rates across sectors of differing organic composition — which Section 5.3 examines directly.

Table 1: Sectoral calibration, Portugal, 2022 (real Eurostat data).

Code	Sector	VAB (M€)	e	k	r (%)
K	Financial and insurance activities	23.2	1.51	3.17	36.1
S	Other service activities	5.0	0.62	1.97	20.83
I	Accommodation and food service	33.8	0.86	3.51	19.07
G	Wholesale and retail trade	54.8	0.53	2.05	17.37
A	Agriculture, forestry and fishing	16.3	1.39	8.3	14.98
N	Administrative and support	16.0	0.33	1.28	14.37
F	Construction	19.8	0.33	1.61	12.65
M	Professional, scientific and technical	21.2	0.34	1.79	12.23
C	Manufacturing	88.0	0.53	3.91	10.75
B	Mining and quarrying	3.2	0.65	9.44	6.19
J	Information and communication	25.1	0.22	2.84	5.71
P	Education	21.6	0.1	1.08	4.8
Q	Human health and social work	34.4	0.1	1.93	3.41
H	Transportation and storage	43.3	0.23	6.33	3.13
L	Real estate activities	382.3	8.52	333.82	2.55
E	Water supply, sewerage, waste	15.6	0.34	13.78	2.3
R	Arts, entertainment and recreation	7.9	0.12	4.15	2.29
O	Public administration	88.5	0.1	7.45	1.18
D	Electricity, gas, steam	33.4	0.1	62.91	0.16

5.2 Aggregate profit-rate trend, 1996–2022

Figure 1 shows the aggregate rate of exploitation, organic composition, and profit rate for Portugal over 1996–2022, calibrated entirely from real AMECO series. The rate of exploitation rises from 0.21 to 0.61; the organic composition falls from 8.99 to 5.40; the profit rate rises correspondingly from 2.1% to 9.5%. We discuss this finding in Section 6.1.

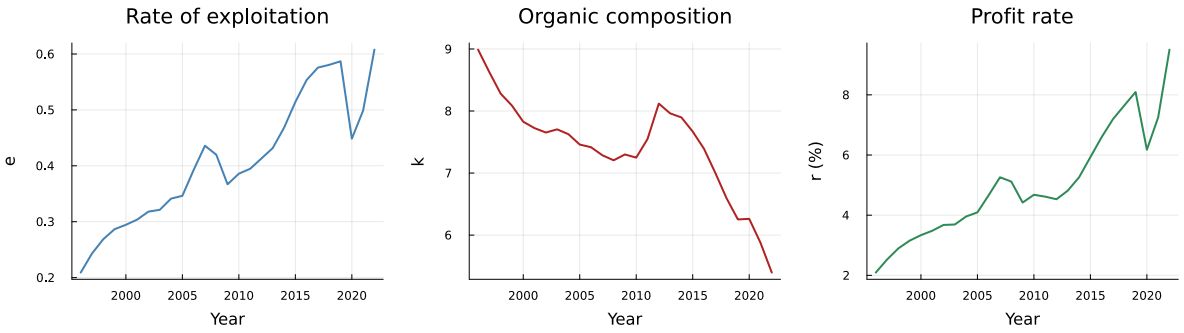


Figure 1: Rate of exploitation, organic composition, and profit rate, Portugal, 1996–2022 (real AMECO data).

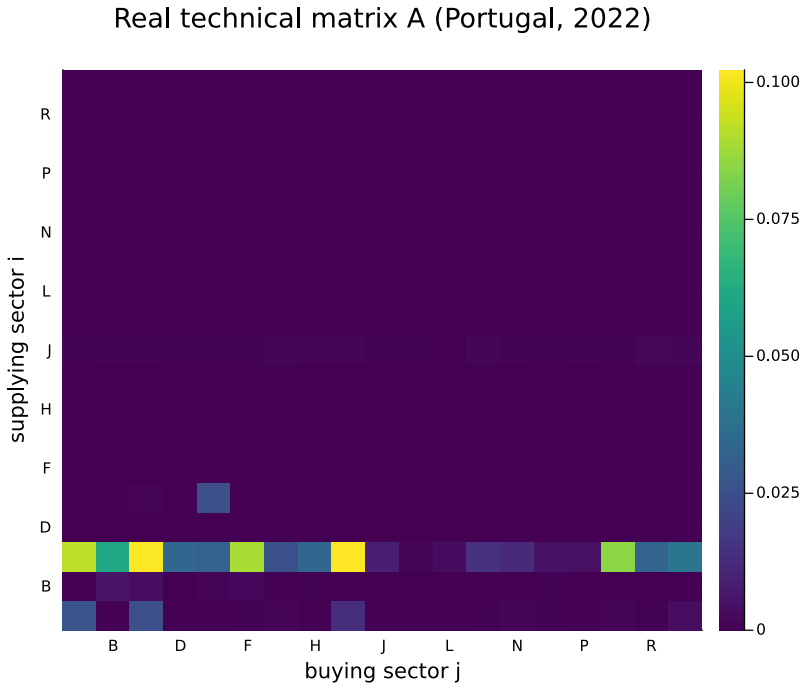


Figure 2: Real technical matrix A, Portugal, 2022 (19 NACE sections).

5.3 The transformation problem on the real economy

Figure 3 compares labor values with Sraffian production prices computed on the real calibrated matrix, for three illustrative uniform profit rates ($\bar{r} = 0$, $0.3R_{\max}$, $0.6R_{\max}$, with $R_{\max} \approx 691\%$ for this economy). As $\bar{r}$ rises, sectors with above-average organic composition see their production price rise relative to their labor value, and vice versa, the qualitative signature the transformation problem predicts, now shown on a real rather than a randomly generated technical structure.



Figure 3: Labor values vs. Sraffian production prices, real technical matrix, Portugal 2022, at three illustrative profit rates.

5.4 Illustrative comparison of coupling mechanisms

Because no real per-sector accumulation time series exists to calibrate the dynamic coupling mechanisms of Section 3.3 empirically, Figure 4 illustrates their qualitative behavior on a synthetic six-sector economy under a common random technical matrix and seed. Capital mobility alone reduces cross-sector profit-rate dispersion over time, as expected. Adding the aggregate realization-pressure term leaves the qualitative picture largely unchanged in this run. Adding exact quantity rationing instead *increases* final dispersion relative to mobility alone in this particular random economy: rationing can prevent the mobility term from fully equalizing rates when a high-profit sector's own suppliers cannot support the investment mobility would otherwise direct there. We report this honestly as an example of the mechanism's behavior rather than evidence that rationing systematically increases or decreases dispersion in general — the result is specific to this synthetic economy's random technical structure, and a systematic characterization across many random economies is left for future work.

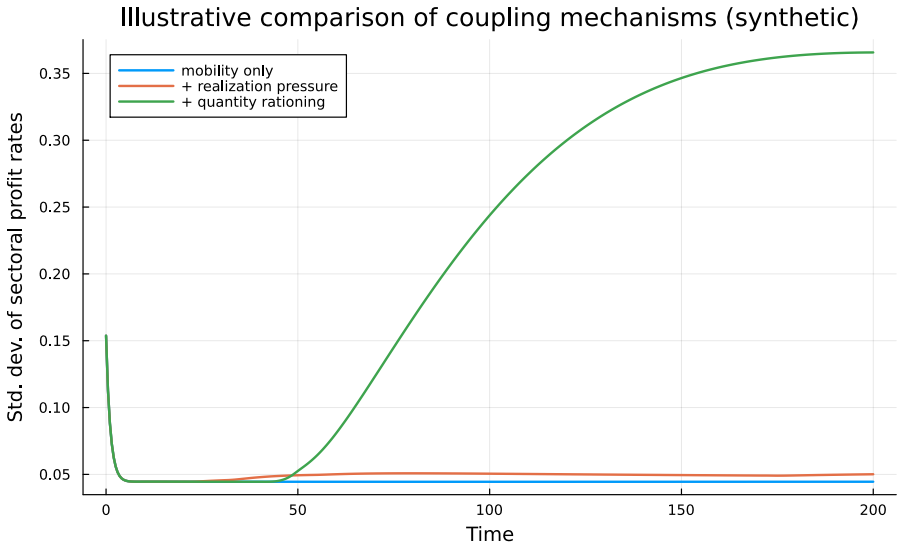


Figure 4: Cross-sector profit-rate dispersion over time under three coupling mechanisms (illustrative synthetic economy, not calibrated to real data).

6 Discussion

6.1 Interpreting the rising profit rate

The rising profit rate found for Portugal over 1996–2022 runs counter to the classical tendential-fall expectation, in which a rising organic composition is expected to depress the profit rate faster than a rising rate of exploitation can offset it. Here the opposite pattern holds: the organic composition *falls* over the period, and the rate of exploitation rises enough on its own to more than compensate. We consider this a genuine empirical finding from unmodified official data, not an artifact of the calibration pipeline, but we do not consider it a settled interpretation of underlying economic processes. Several non-exclusive candidate explanations are worth further investigation: intensified exploitation associated with wage moderation and labor-market liberalization over the period, including during the 2011–2014 assistance programme; capital-saving technical change, plausible given the economy's structural shift toward less capital-intensive services; and compositional effects from Portugal's deepening integration into EU-wide production networks. Distinguishing between these is beyond the scope of this paper and would require disaggregating the aggregate series by sector and comparing against the 2022 cross-section of Table 1 — itself complicated by the single-year nature of the sectoral calibration versus the time-series nature of the aggregate result.

6.2 Limitations

We report limitations transparently, following the same practice as the package's own documentation.

Default behavior remains synthetic. Both calibration paths default to synthetic data or fixed sectoral weights; every real-data result in this paper required explicit opt-in flags. This is a deliberate design choice to keep the package's automated test suite fully offline, not a limitation of the real-data pipeline itself, but it means synthetic results should never be mistaken for empirical ones when using the package's defaults.

Single-rate depreciation. The perpetual-inventory capital-stock option (Section 4.3) uses one depreciation rate per sector, not differentiated by asset type. Eurostat publishes investment by asset type for the relevant dataflow, but not the corresponding asset-specific depreciation profiles, which would be needed for a fully differentiated perpetual-inventory calculation.

No microfoundation for final demand. The quantity-rationing mechanism of Section 3.3 constrains reinvestment demand through the technical matrix but does not model how aggregate consumption demand is allocated across sectors, since no data source in the current pipeline provides sector-of-origin consumption baskets. The mechanism should therefore be read as constraining the investment-goods side of reproduction specifically, not final demand in general.

Coarser sectoral aggregations inherit a data-quality caveat. When the technical matrix is aggregated from 19 to 3 macro-sectors, the tertiary (services) row is conservative because it inherits reduced intermediate-flow estimates for the trade and transport sub-sectors, whose margin-accounting convention we clip at the finer resolution before aggregating (Section 4.2); the coarser matrix is non-negative and usable, but likely understates services' true role as an input supplier to other sectors.

7 Conclusion

MarxKapital demonstrates that a real, live, officially sourced data pipeline for Marxian value-theoretic calibration is practical using current European statistical infrastructure, and that exact per-sector market-clearing dynamics can be added to a reproduction-schema model without the numerical overhead of a formal differential-algebraic formulation. Applied to Portugal, the pipeline surfaces a genuine and, we think, under-examined empirical pattern — a rising profit

rate driven by rising exploitation outweighing falling organic composition — that merits further sectoral and comparative investigation beyond the scope of this paper.

Availability

MarxKapital is released under the MIT license. The complete source code, including the empirical pipeline described here, is available at https://github.com/randsley/MarxKapital. Every figure and table in this paper is produced, without manual editing, by `paper/gerar_resultados.jl`, which performs live queries against the Eurostat and AMECO SDMX APIs; re-running it reproduces this paper's results from the same official data sources.

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